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The Idea of the Universe as a Black Hole Revisited

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Abstract—The Friedmann equation for a homogeneous and isotropic universe model with a cosmological constant yields the eventual maximal expansion of a closed dust universe. On the other hand, from the line element for the false vacuum outside a spherically matter distribution we obtain the generalized Schwarzschild radius, i.e., the position of the event horizon. It has been argued that the mathematical similarity of these two quantities cannot be a coincidence, but means that the universe is inside a black hole. We show that this interpretation is wrong. Matching the two metrics, we find what we intuitively expect: our expanding universe cannot be inside the event horizon of the vacuum metric.

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1. INTRODUCTION

Einstein's theory for gravitation, general relativity, showed its eminence predicting that a static universe cannot exist. Hubble's most remarkable discovery confirmed this revolutionary new world picture: There exists a linear relation between the redshift of the light received from distant nebulae and the distance to those galaxies. The accepted proper interpretation of Hubble's law is that space itself is expanding and carrying the nebulae with it. The simplest universe models are obtained accepting a philosophical assumption: The universe should present the same aspect from every point except for local irregularities. This so-called cosmological principle thus accepts that the universe is homogeneous and isotropic.

The line element for space-time is then derived from symmetry arguments alone:

$$ds_-^2 = g^-_{\alpha\beta} dx_-^\alpha dx_-^\beta = -c^2 dt^2 + a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right), \qquad (1)$$

where c is the speed of light in vacuum and t is the proper time of every observer moving with the substratum. The comoving radial space coordinate r and the usual angular coordinates θ and ϕ are dimensionless while the scale factor $a(t)$ has dimension of length. The constant k is $1, 0, -1$ according to whether the spatial geometry of the model has positive, zero or negative curvature.

Moreover, when the universe models are constructed, matter is smoothed out to form a continuous gas characterized by a density ρ and a pressure p. A perfect fluid and the cosmological principle have the great merit of keeping the mathematics simple. Einstein's field equations then yield several possible models for the universe. The Friedmann equation reads

$$8\pi G\rho = 3\frac{\dot{a}^2}{a^2} + 3\frac{kc^2}{a^2} - \Lambda c^2, \qquad (2)$$

where G is Newton's gravitational constant and a dot denotes d/dt. Λ is here the so-called cosmological constant which is now often attributed to a strange form of vacuum energy (also termed dark energy). The universe also obeys the principle that there is no entropy change in adiabatic expansion,

$$\frac{d}{dt}(\rho a^3) + \frac{p}{c^2}\frac{d}{dt}(a^3) = 0. \qquad (3)$$

To solve these equations for the time evolution of the universe, i.e., obtain $a(t)$, one also need to prescribe an equation of state $p(\rho)$. What values to choose for Λ and for the signature $k = \pm 1, 0$ is just a matter of philosophical taste. In our opinion, one should be open-minded. One should work out all possibilities and let comparison with observations decide which model to prefer. However, some time ago Pathria [1] declared that there is a good reason to choose $k = 1$ (closed spatial geometry) a priori. He concluded that this is the superior choice when he compared the Friedmann equation for a dust-filled universe ($p = 0$) and the metric for a false vacuum outside a spherically symmetric matter distribution. What was really sensational and astonishing was his idea that *our universe itself is a black hole*. He claimed that outside our universe there must exist an *external*

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world from which one may take a "detached" look at our universe. This is a highly unusual view to say the least. If true, it could do away with the trouble that one has to imagine a closed 3-dimensional world. Then one could avoid having to insist that mathematics is superior since it can give us an understanding of what is going on even if we cannot make a proper picture of the situation.

2. OUTLINE OF PATHRIA'S INVENTION

From Eq. (3) Pathria observes, that for a universe filled with dust ($p = 0$), it must be the case that

$$C \equiv \frac{8\pi G}{3c^2}\rho a^3 = \text{const.} \tag{4}$$

The Friedmann equation for a dust universe with a closed spatial geometry may then be written as

$$\frac{1}{c^2}\dot{a}^2 = \frac{1}{3}\Lambda a^2 + \frac{C}{a} - 1. \tag{5}$$

At the time when Pathria wrote his paper, observations seemed to tell that the rate of expansion was in decrease. From Eq. (5) it then follows that

$$\frac{2}{3}\Lambda a_0 - \frac{C}{a_0^2} < 0, \tag{6}$$

where a_0 denotes the present-day value of $a(t)$.

Next, Pathria takes a look at the vacuum line element outside a spherical matter distribution:

$$\begin{aligned} ds_+^2 = g^+_{\alpha\beta}dx^\alpha_+ dx^\beta_+ = -\left(1 - \frac{2m}{R} - \frac{1}{3}\Lambda R^2\right)c^2 dT^2 \\ + \left(1 - \frac{2m}{R} - \frac{1}{3}\Lambda R^2\right)^{-1} dR^2 \\ + R^2(d\theta^2 + \sin^2\theta d\phi^2), \end{aligned} \tag{7}$$

where R is the radial coordinate, T is the time coordinate, and $m = \text{const}$ denotes the geometric mass.

It should be remembered that Birkhoff's theorem is also valid when a cosmological constant Λ is included in the field equations [2]. Hence, any spherically symmetric distribution of matter, even if in radial motion, leads to the *same* exterior line element to the matter distribution.

Pathria claims that m is given by the following expression:

$$m = \frac{4\pi G}{c^2}\int_0^a \rho(R)R^2 dR = \frac{4\pi G}{3c^2}\rho a^3. \tag{8}$$

He is then struck by the mathematical similarity for the right-hand side of Eq. (5) and the metric coefficient

$$g^+_{44} = \frac{1}{3}\Lambda R^2 + \frac{2m}{R} - 1 \tag{9}$$

of the line element for the vacuum.

From Eq. (7) it is seen that T ceases to be a time marker if $g^+_{44} > 0$. In this case the matter sphere is inside its own event horizon (Schwarzschild surface). Hence, the matter distribution is certainly inside a black hole if we have $R_{\text{surface}} < R_s$ where R_s is the Schwarzschild radius, i.e., the least positive solution of the equation $g^+_{44} = 0$ and if we also have

$$\frac{\partial}{\partial R}g^+_{44}(R = R_s) = \frac{2}{3}\Lambda R_s - \frac{2m}{R_s^2} < 0. \tag{10}$$

Note that this last condition yields a positive temperature of the Hawking radiation from vacuum just outside the event horizon. An excellent discussion of this topic is given by Taylor and Wheeler [3]. Equation (10) also reminds us of Eq. (6) concerning deceleration of the universe expansion. The Schwarzschild radius R_s obtained from the exterior metric (7) is declared to represent what is called the Schwarzschild radius for the universe itself. From Eq. (5) it is then immediately seen that the maximal value $a_{\max}$ of the scale factor $a(t)$ is equal to R_s.

3. SOME PRELIMINARY REMARKS CONCERNING PATHRIA'S APPROACH

We first want to say that Pathria is occasionally employing a dangerous notations. First, he denotes the time coordinate in the Friedmann-Lemaître-Robertson-Walker (FLRW) metric (1) and the time coordinate in the exterior vacuum metric (7) by the same symbol t. We have changed this bad notation. The latter time coordinate is denoted by T.

Second, we have changed the notation for the scale factor in the FLRW metric to $a(t)$ to avoid mixing of concepts found in Eq. (8) where Pathria calculates the geometric mass of the fluid sphere. In this calculation it seems that the radial coordinate in the exterior vacuum metric is identified with the scale factor. To escape this confusion, we denote the comoving radial coordinate in the FLRW-metric by small r and the the radial coordinate in the vacuum metric by capital R. The following equation then tells us at which external radial coordinate a particular layer of matter is at given time,

$$R = a(t)r. \tag{11}$$

The surface of a fluid sphere is thus given by

$$R_b = a(t)r_b, \tag{12}$$

where the index b means boundary. The geometric mass of a fluid sphere is

$$m = \frac{4\pi G}{c^2}\int_0^{R_b} \rho\overline{R}^2 d\overline{R}$$

$$= \frac{4\pi G}{c^2} \int_0^{a(t)r_b} \rho(t)[ra(t)]^2 d(ar)$$

$$= \frac{4\pi G}{c^2} \int_0^{r_b} \rho(t) a^3(t) r^2 dr$$

$$= \frac{4\pi G}{c^2} \rho a^3 \int_0^{r_b} r^2 dr = \frac{4\pi G}{3c^2} \rho a^3 {r_b}^3. \quad (13)$$

Hence, the expressions (8) and (13) for the geometric mass m are the same only if the radial comoving coordinate r_b takes the value unity.

The expression (13) for m can be checked for consistency using the so-called mass function $m(r,t)$. Different expressions for it are found in the literature [4–8]. Here, however, it is most convenient to follow the paper by Eisenstaedt [9] where the cosmological constant is included. The most general spherically symmetric metric may be written as

$$ds^2 = -e^{2\nu(u,\tau)} d\tau^2 + e^{2\lambda(u,\tau)} du^2 + e^{2\psi(u,\tau)}(d\theta^2 + \sin^2\theta d\phi^2). \quad (14)$$

The time and radial coordinates are here denoted by τ and u, respectively. Units are chosen so that $G = c = 1$. The mass function $m(u,\tau)$ is then given by

$$m = \frac{1}{2} e^{\psi} \left[1 + g^{ij} \left(\frac{\partial e^{\psi}}{\partial x^i} \right) \left(\frac{\partial e^{\psi}}{\partial x^j} \right) - \frac{1}{3} \Lambda e^{2\psi} \right]. \quad (15)$$

Using the Friedmann equation (2) for the FLRW metric, we obtain the following expression for the mass function:

$$m(r,t) = \frac{4\pi G}{3c^2} \rho a^3 r^3, \quad (16)$$

which shows consistency with the geometric mass m if we put $r = r_b$.

Now, we want to remark that we have no objections if the FLRW metric is employed to examine what happens to a homogeneous and isotropic contracting fluid sphere or matter distribution expanding into a false vacuum if the condition $R_b > R_s$ is fulfilled. What we cannot accept is the interpretation that such a fluid sphere with $a_0 r_b < R_b$ can represent our universe expanding into an external world. If the surface of the fluid sphere is inside its own event horizon, all light cones in the vacuum region between this surface and the Schwarzschild surface point inward. Hence, a particle at the surface has no choice but to move inward, which must also be the case for the fluid sphere itself.

Let us temporarily forget this obvious contradiction and follow Pathria's idea seriously, taking the confused conception of an external world for a closed universe to represent more than a fictitious fantasy. We shall match the FLRW metric with that of a false vacuum to see where this leads us.

4. JUNCTION CONDITIONS

We want to obtain the sufficient and necessary conditions for smooth matching across the surface Σ of the interior region V^- with a perfect fluid and the vacuum exterior V^+. The FLRW metric is valid in the interior region, and the exterior metric is given by Eq. (7). We find it most convenient to follow the procedure given by Santos [10]. This procedure was also followed by Bonnor et al. [11].

The surface Σ, the boundary of the interior fluid sphere, will have the following equations in V^- and V^+, respectively:

$$f(x_i) = r - r_b = 0, \quad (17)$$

$$\overline{f}(\overline{x_i}) = R - R_b(T) = 0. \quad (18)$$

Using (17) in (1), we have for the metric on Σ

$$ds_\Sigma^2 = -c^2 dt^2 + a^2 r_b^2 (d\theta^2 + \sin^2\theta d\phi^2). \quad (19)$$

It is useful to write

$$A(t) \equiv a(t) r_b. \quad (20)$$

Next, we use (18) in (7) and find that on Σ Eq. (7) reduces to

$$ds_{\Sigma}{}^2 = -\Bigg[1 - \frac{2m}{R_b} - \frac{1}{3}\Lambda R_b^2 - \frac{1}{c^2}\left(\frac{dR_b}{dT}\right)^2 \left(1 - \frac{2m}{R_b} - \frac{1}{3}\Lambda R_b^2\right)^{-1} \Bigg] c^2 dT^2 + R_b^2 \left(d\theta^2 + \sin^2 d\phi^2\right). \quad (21)$$

The first fundamental form which Σ inherits from the interior metric must be the same as the one it inherits from the exterior metric. This will be the case if the following two conditions hold:

$$A(t) = a(t) r_b = R_b(T), \quad (22)$$

$$\frac{dT}{dt} = \Bigg[1 - \frac{2m}{R_b} - \frac{1}{3}\Lambda R_b^2 - \frac{1}{c^2}\left(\frac{dR_b}{dT}\right)^2 \left(1 - \frac{2m}{R_b} - \frac{1}{3}\Lambda R_b^2\right)^{-1} \Bigg]^{-1/2}, \quad (23)$$

where we assume that

$$1 - \frac{2m}{R_b} - \frac{1}{3}\Lambda R_b^2 - \frac{1}{c^2}\left(\frac{dR_b}{dT}\right)^2 \left(1 - \frac{2m}{R_b} - \frac{1}{3}\Lambda R_b^2\right)^{-1} > 0, \quad (24)$$

so that T is a timelike coordinate.

Next we examine the conditions that the second fundamental forms inherited from both metrics must also be the same on Σ. We need the outward unit normals n_i to Σ both in V^- and V^+. The general expression for unit normals as given by Bonnor and Vickers [12] reads

$$n_i = \frac{\partial f}{\partial x^i}\left(g^{\alpha\beta}\frac{\partial f}{\partial x^\alpha}\frac{\partial f}{\partial x^\beta}\right)^{-1/2}. \tag{25}$$

Recalling Eqs. (17) and (18) and using (23), we obtain

$$n_i^- = \left(0, \frac{a(t)}{\sqrt{1-r^2}}, 0, 0\right), \tag{26}$$

$$n_i^+ = (-\dot{R}_b, \dot{T}, 0, 0). \tag{27}$$

We now calculate the intrinsic and extrinsic curvature $K_{ij}^{\pm}$ on Σ. These curvatures are given by [10]

$$K_{ij}^{\pm} = -n_i^-\frac{\partial^2 x_\pm^\alpha}{\partial\xi^i\partial\xi^j} - n_\alpha^-\Gamma^\alpha_{\mu\nu}\frac{\partial x_\pm^\mu}{\partial\xi^i}\frac{\partial x_\pm^\nu}{\partial\xi^j}, \tag{28}$$

where $\Gamma^\alpha_{\mu\nu}$ are the Christoffel symbols. We take $\xi^0 = t$, $\xi^2 = \theta$, and $\xi^3 = \phi$ to be the parameters on Σ. We will also drop the index b: all quantities are to assume their values on the surface Σ.

A straightforward calculation now yields

$$K_{\theta\theta}^- = r\sqrt{1-r^2}\,a(t), \tag{29}$$

$$K_{\theta\theta}^+ = \dot{T}R\left(1 - \frac{2m}{R} - \frac{1}{3}\Lambda R^2\right). \tag{30}$$

From now on we find it useful to write

$$B \equiv 1 - \frac{2m}{R} - \frac{1}{3}\Lambda R^2. \tag{31}$$

We also immediately obtain

$$K_{tt}^- = 0. \tag{32}$$

After a somewhat tedious calculation we find that K_{tt}^+ is given by

$$K_{tt}^+ = \dot{A}\ddot{T} - \dot{T}\ddot{R} + \left(\frac{m}{R^2} - \frac{1}{3}\Lambda R\right) \times \left(2\frac{\dot{A}}{B}\dot{T}\dot{R} - c^2B\dot{T}^3 + \frac{1}{B}\dot{T}\dot{R}\right). \tag{33}$$

We want to remove c^2B from this expression. Recalling Eqs. (22) and (23), we obtain

$$c^2B = \frac{\dot{A}^2}{B\dot{T}^2 - 1}. \tag{34}$$

Inserted into Eq. (33), this yields

$$K_{tt}^+ = \dot{R}\ddot{T} - \dot{T}\ddot{R} + \left(\frac{m}{R^2} - \frac{1}{3}\Lambda R\right)\frac{\dot{T}\dot{R}^2(2B\dot{T}^2 - 3)}{B(B\dot{T}^2 - 1)}. \tag{35}$$

Now we remove $\ddot{T}$ from the last equation, using (29) and (30). We must have $K_{\theta\theta}^- = K_{\theta\theta}^+$. Hence, we can write

$$\dot{T}B = \sqrt{1-r^2}. \tag{36}$$

We differentiate and obtain

$$\ddot{T} = -2\dot{T}\dot{R}D/B, \tag{37}$$

where

$$D = \frac{m}{\dot{R}^2} - \frac{1}{3}\Lambda R. \tag{38}$$

Thus the extrinsic curvature is given by

$$K_{tt}^+ = -\frac{D\dot{T}\dot{R}^2}{B(B\dot{T}^2 - 1)} - \dot{T}\ddot{R}. \tag{39}$$

We must have $K_{tt}^- = K_{tt}^+$. Equations (32) and (39) then yield

$$\ddot{R} = -c^2D, \tag{40}$$

where we have taken Eq. (34) into account. We are now ready to examine what is the physical contents of the condition (40).

The FLRW metric for a perfect fluid also yields

$$8\pi G\frac{p}{c^2} = -2\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} - \frac{kc^2}{a^2} + \Lambda c^2. \tag{41}$$

We now take $k = 1$ and use of Eq. (22) and the condition (40) to calculate the pressure at the surface. We find

$$8\pi G\frac{p}{c^2} = \frac{2Dc^2}{R} - \frac{\dot{R}^2}{R^2} - \frac{r_b^2c^2}{R^2} + \Lambda c^2. \tag{42}$$

Now, once more we take into account the condition $K_{\theta\theta}^- = K_{\theta\theta}^+$ and employ Eq. (36). We find what we expected. The right-hand side of Eq. (42) simplifies beautifully, and we conclude: *The interior region with an FLRW metric and a perfect fluid is matched smoothly with the exterior false vacuum only if the pressure vanishes at the surface.*

Recalling that the pressure is a function of time only, we conclude that the pressure is zero identically. Hence, the fluid sphere is a dust sphere.

From our procedure of matching the two different regions it is also easily seen that our formula (13) for the geometric mass m of the fluid sphere is confirmed. Our condition $K_{\theta\theta}^- = K_{\theta\theta}^+$ and Eq. (34) immediately yield

$$m = \frac{r_b^3\dot{a}^2a}{2c^2} + \frac{r_b^3a}{2} - \frac{\Lambda r_b^3a^3}{6}. \tag{43}$$

Recalling the Friedmann equation (2), our expression (13) for the geometric mass is obtained.

Eisenstaedt's formula (15) for the mass function is also easily confirmed. When the mass function is calculated using the FLRW metric and Eq. (15), Eq. (42) is obtained.

5. THE CHOICE $r_b = 1$

With the condition $K^-_{\theta\theta} = K^+_{\theta\theta}$ at the boundary, it is now seen from Eqs. (29) and (30) that Pathria's choice $r_b = 1$ yields

$$1 - \frac{2m}{R_b(T)} - \frac{1}{3}\Lambda R_b^2(T) = 0. \qquad (44)$$

Hence, this choice of boundary value for the comoving radial coordinate just means that the fluid sphere is *static*, and the surface of the fluid sphere merges with its event horizon. Thus this particular model cannot describe our expanding universe. Moreover, we observe that the FLRW metric (1) is singular if $r_b = 1$. We can avoid this singularity using the coordinate transformation

$$r = \sin\psi. \qquad (45)$$

The FLRW metric then reads

$$ds^2 = -c^2dt^2 + a^2d\psi^2 + a^2\sin^2\psi d\theta^2 + a^2\sin^2\psi\sin^2\theta d\phi^2. \qquad (46)$$

The area F of the surface of the fluid sphere characterized by a radial coordinate ψ is given by

$$F = 4\pi a^2\sin^2\psi. \qquad (47)$$

This area increases with ψ as expected for $\psi \in [0, \pi/2]$. This area takes its maximum value for $\psi = \frac{\pi}{2}$, i.e., $r_b = 1$. But what is enclosed within this maximal surface is just half of the universe. Exactly here the very different concepts of an isolated fluid sphere surrounded by empty space and a closed Friedmann universe part. The trouble is that we cannot imagine a closed universe. But the strange geometry still exists. The area F decreases when $\psi \in (\pi/2, \pi]$ and disappears completely when $\psi = \pi$. There simply does not exist a firm ground outside a closed universe where a distant observer can stand if he wants to measure the geometric mass. Moreover, the geometric mass $m = \frac{4}{3}\pi\frac{G}{c^2}\rho a^3\sin^3\psi$ will vanish for this closed universe. Hence, for this universe model one can say that the total mass of all particles is canceled by the mass defect. In our opinion, it is better to say that the very concept is meaningless.

Our final conclusion is that Pathria's interpretation of some mathematical similarities certainly served as an inspiration for a guesswork that is much too drastic. A declaration is no explanation. A closer look shows that the model just represents careless notation and confused concepts.

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